Nethuni Rajapakse

Software Engineer

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Education

University of Moratuwa — Moratuwa, SL
B.Sc. (Hons) in Information Technology — *Jun. 2022 – present*

- Expected Graduation: Jun. 2026
- CGPA (up to L3): 3.65 / 4.0

Southlands College — Galle, SL
G.C.E. Advanced Level Examination (2020) — *Jan. 2011 – Jan. 2021*

- Physical Stream
- Z-score: 1.7603

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL (PostgreSQL, MySQL), C, HTML/CSS
Frameworks: Spring Boot, React, FastAPI, QueryDSL, WebSockets
Developer Tools: Git, Docker, Postman, Figma, VS Code, IntelliJ, Selenium
Data & AI: PostgreSQL, MySQL, Redis, Apache Kafka, XGBoost, SHAP
Concepts: Event-Driven Microservices, REST APIs, JWT Authentication, Agile/Kanban

Professional Experience

Software Engineering Intern — Feb. 2025 – Aug. 2025
Revocare Solutions — *Colombo, SL*

- Maintained a large-scale FinTech platform using **Spring Boot** and **React** in a **Kanban**-driven team.
- Optimized frontend data fetching by implementing **TanStack Query** and improving custom hook performance.
- Improved backend performance using **QueryDSL** and batch-fetching to reduce SQL overhead.
- Built unit and integration tests with **JUnit** and **Mockito** to validate core APIs.
- Developed modular CRM components and dynamic filters to improve financial data visualization.
- Implemented health checks and rate limiting to enhance microservice reliability and security.

Final Year Research Project

Interlekt – Hallucination Mitigation in Sinhala Small Language Models | GitHub — 2026

- Investigating hallucination mitigation for fact-sensitive Sinhala question answering under the combined constraints of low-resource language, small language model, and on-device deployment, supervised by *Dr. Thilina Thanthriwatte*, Faculty of IT, University of Moratuwa.
- Leading the Edge Deployment & Quantisation Study: building an on-device inference pipeline that integrates **llama.cpp** via the **Android NDK** with a **JNI** bridge to a **Kotlin/Jetpack Compose** layer, supporting runtime loading of any **GGUF** quantisation variant (Q3/Q4/Q5/Q8).
- Designing the quantisation impact evaluation: a 50-question Sinhala QA set across 8 experimental conditions (4 quantisation levels × 2 RAG conditions), measuring factual accuracy, hallucination rate, and on-device hardware feasibility.

Projects

Rostra – Real-Time Auction Platform | *Spring Boot, Kafka, PostgreSQL, WebSockets, Docker* | GitHub — 2026 (Ongoing)

- Designing an **event-driven microservices** system with four independently deployed **Spring Boot** services (Auth, Auction, Bidding, Notification), each owning a dedicated **PostgreSQL** database.
- Implementing the **Transactional Outbox pattern** to guarantee at-least-once **Kafka** event delivery alongside database commits, eliminating dual-write inconsistency.
- Applying **optimistic locking** with JPA `@Version` to safely resolve concurrent bid races without distributed transactions.
- Securing inter-service and client communication with **JWT**-based authentication and a **Spring Security** filter chain at the API Gateway.

- Streaming live auction state to clients via **WebSockets** (STOMP) and integrating **Resilience4J** circuit breakers to prevent cascading failure between services.

EstateVision – Sri Lanka Property Price Predictor | *Python, XGBoost, SHAP, Flask, React* | GitHub 2026

- Built an end-to-end ML pipeline predicting Sri Lankan property prices, achieving **R^2 of 0.71** and **MAE of Rs. 13.4M** on a 614-listing held-out test set.
- Scraped 618 property listings across 15 districts from ikman.lk using **Selenium** with checkpoint-based fault tolerance and anti-detection measures.
- Engineered 20 features including geographic tiering, Colombo premium flags, and 10 amenity signals extracted from listing descriptions.
- Tuned an **XGBoost** regressor using **RandomizedSearchCV** (30 candidates, 5-fold CV) with log-transformed targets to handle right-skewed price distribution.
- Integrated **SHAP TreeExplainer** for per-prediction explanations, served via a **Flask** REST API and visualized in a **React/TypeScript** frontend.

SpotiInsights | *Python, FastAPI, Spotify Web API, PostgreSQL, Redis* | GitHub 2025 (Ongoing)

- Developing a data-driven application to visualize Spotify listening patterns and audio features.
- Integrated the **Spotify Web API** with OAuth authentication to securely access user data.
- Designed RESTful APIs using **FastAPI** for efficient data retrieval and processing.
- Implemented caching with **Redis** to optimize performance and reduce external API calls.

Ascendia – Construction Project Management Tool | *Spring Boot, MySQL, React.js* | GitHub L2 Software Project

- Developed a role-based project management system for large-scale construction workflows supporting five distinct user roles.
- Implemented the Project Manager module with task and job management features.
- Integrated email notifications for project and task assignments.
- Collaborated in a team environment as a **Full-Stack Developer**.

Achievements & Participations

- Cre8X Designathon (KDU) – **Top 10 Finalists**
- CodeRush 2023 (UOM INTECS) – **Top 30**
- ACES Coders 2024 (UOP) – Participant
- PyHack 2.0 (UOK) – Participant
- MoraXtreme 8.0 – Participant

Volunteering & Leadership

- **IEEE Student Branch of UOM:** Mora Foresight 1.0 (Finance Committee); VR Cade (Finance Committee); HackElite (Organizing Committee).
- **Rotaract Club of UOM:** Rotaract Career Fair (Company Coordinator); Nena Aruna (Volunteer Tutor – Mathematics).
- **LEO Club of UOM:** Future Wave (Volunteer Tutor – Arduino).
- **Classical Music Society of UOM:** Yathra Annual Concert (Organizing Committee).
- **UOM INTECS:** CodeRush 2025 (Finance Committee).

References

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